EE Department, Sharif University of Technology

Blind Separation of Nonlinear Mixtures of Stochastic Processes

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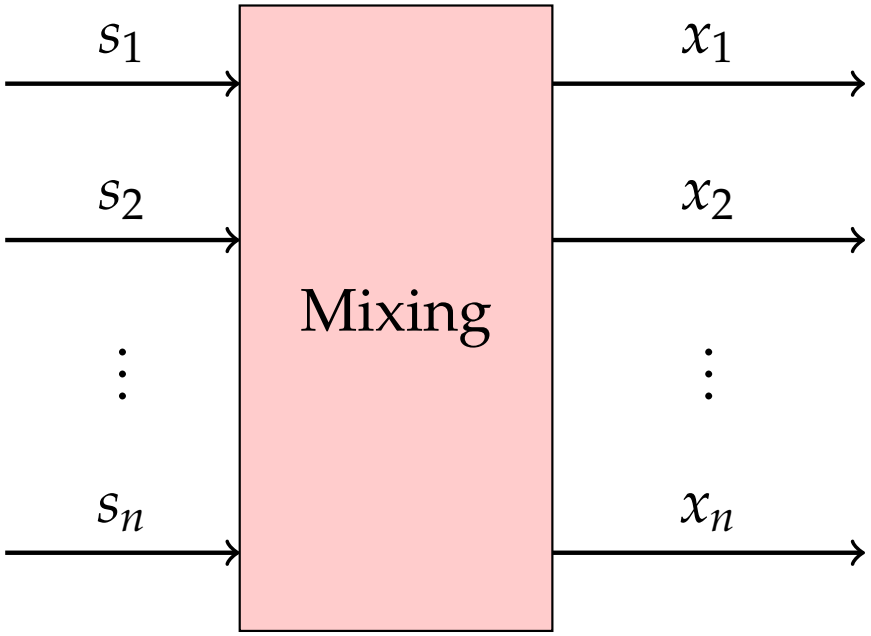
Outline

1. Blind Source Separation
2. Linear and Non-Linear BSS
3. Using Time Structure of the Data
4. New Separability Results
5. This Project
6. Bibliography

Blind Source Separation
Introduction



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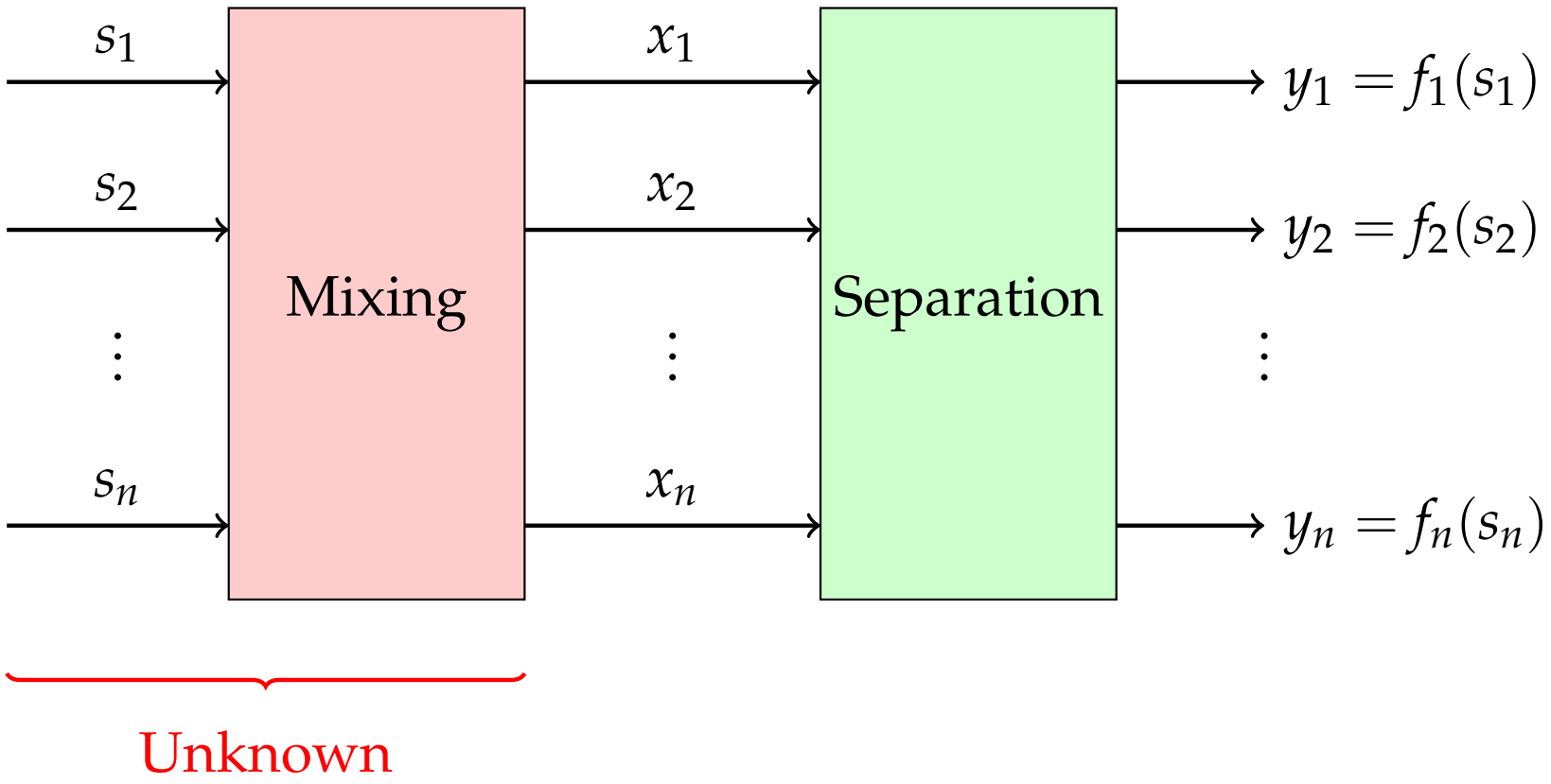

Linear and Non-Linear BSS

Darmois-Skitovic Theorem [Darmois-Skitovich 1950]

In the linear setting, the model is identifiable if the sources are **non-Gaussian** random variables.

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Non-Linear Mixtures

Non-linear mixtures are harder!

$$\begin{cases} S_1 = \text{Rayleigh}(\sigma) \\ S_2 = \text{Uniform}[0, 2\pi] \end{cases} \implies X_1 = S_1 \cos(S_2) \perp\!\!\!\perp X_2 = S_1 \sin(S_2)$$

Using Time Structure of the Data

Time relations can help! [Hosseini et al., 2003]

The output **Stochastic Processes** are not independent.

$$
\begin{aligned}
R_{x_1 x_2}(t, t+1) &= \mathbb{E}\big[x_1(t+1)x_2(t)\big] \\
&= \mathbb{E}\Big[s_1(t+1)s_1(t)\Big]\mathbb{E}\Big[\cos\big(s_2(t+1)\big)\sin\big(s_2(t)\big)\Big]
\end{aligned}
$$

New Separability Results

Permutation Contrastive Separation [Hyvarinen et al., 2017]

The mixture $\mathbf{x}(t) = \mathbf{f}\big(\mathbf{s}(t)\big)$ is separable if:

- $\mathbf{f}$ is invertible and smooth!
- $s_i(t)$: **stationary**, **ergodic** and ***uniformly dependent***.
- $s_i(t)$ are not ***quasi-Newton***.

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Time Contrastive Learning [Hyvarinen et al., 2016]

The mixture $\mathbf{x}(t) = \mathbf{f}\big(\mathbf{s}(t)\big)$ is separable if:

- $\log p_\tau(s_i) = \lambda_i(\tau) q(s_i) - \log Z(\lambda_i(\tau))$
- $[\mathbf{L}]_{\tau,i} = \lambda_{i,1}(\tau) - \lambda_{i,1}(1)$ has full column rank.

This Project

Generalized TCL

Generalize TCL for

$$\log p_\tau(s_i) = q_{i,0}(s_i) + \sum_{v=1}^{V} \lambda_{i,v}(\tau) q_{i,v}(s_i) - \log Z(\lambda_{i,1}(\tau), \ldots, \lambda_{i,v}(\tau))$$

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Sparse Signals

If the signal is sparse in time, BSS is easy! What if it is sparse in some other domain?

Separation Algorithms

Generalizing Information Minimization Algorithms for Non-Linear BSS. [Babaie-Zadeh et al., 2005]

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Final Goal (?)

A General Indentifiability Result for Smooth Mixtures. Is Smoothness Enough

Idea: Nonlinear BSS is linear in an specific high dimensional feature space.

Selected Papers I

[1] **M. Babaie-Zadeh and Ch. Jutten.**
A general approach for mutual information minimization and its application to blind source separation.
Signal Processing, 85(5):975–995, 2005.

[2] **Shahram Hosseini and Christian Jutten.**
On the separability of nonlinear mixtures of temporally correlated sources.
IEEE signal processing letters, 10(2):43–46, 2003.

[3] **Aapo Hyvarinen and Hiroshi Morioka.**
Unsupervised feature extraction by time-contrastive learning and nonlinear ica.
Advances in Neural Information Processing Systems, pages 3765–3773, 2016.

[4] **Aapo Hyvarinen and Hiroshi Morioka.**
Nonlinear ICA of temporally dependent stationary sources.
20th International Conference on Artificial Intelligence and Statistics (AISTAT), 2017.

[5] **Ch. Jutten, M. Babaie-Zadeh, and J. Karhunen.**
Nonlinear Mixtures in Handbook of Blind Source Separation: Independent component analysis and applications.
Academic press, 2010.

Thank You!